SIMATS ENGINEERING

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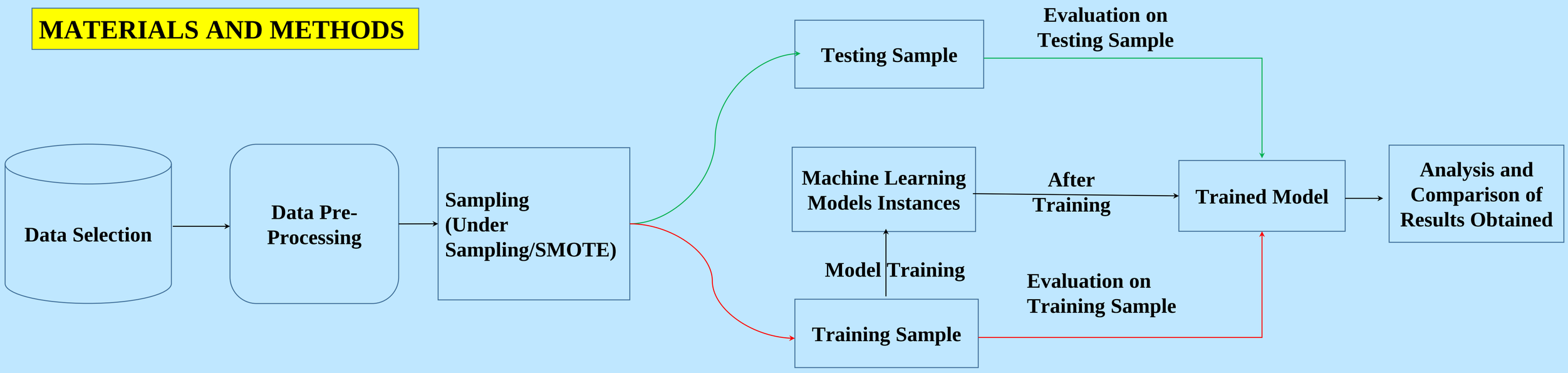
Fraudulent Scam Call Detection with Improved Accuracy using KNN Algorithm Compared with Decision Tree Algorithm.

INTRODUCTION

- The objective of this project is to enhance the accuracy of a fraudulent scam call detection , by comparing the performance of KNN algorithm with a Decision Tree algorithm.
- Scam calls may seek to extract sensitive data, including personally identifiable information (PII). Scam calls can also target businesses, attempting to trick employees or compromise sensitive corporate information. Fraudulent scam call detection initiatives contribute to public awareness and education.
- The application of fraudulent scam call detection are Telecommunications, Financial Institutions, Ecommerce Platforms, Government Agencies, Health Care and Insurance Companies .
- The advantage of using KNN algorithm has given better accuracy compared with Decision Tree algorithm.
- The dataset used was the CDR-Call-Details dataset obtained from Kaggle, consisting of 25,000 call details records and with a size of 2 MB. The dataset was divided into a training set of 19,330 details and a test set of 5,670 details.

Fig 1. Detect and Block Scam Calls

MATERIALS AND METHODS



Fraudulent Scam Call Detection Flow Diagram

RESULTS

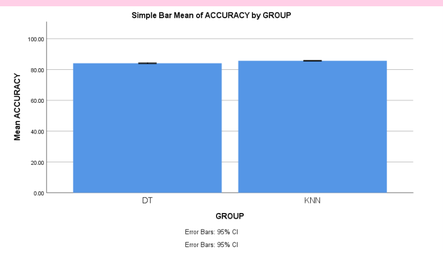


Fig 2. Accuracy value comparison of KNN and DT Algorithm.

Table 1. Mean, Standard Deviation, Standard Error Mean with an accuracy rate comparison of KNN over DT Algorithm .

Group Statistics				
Group Name	N	Mean	Standard Deviation	Std. Error Mean
KNN	20	85.6632	.45158	.10098
Decision Tree	20	84.0576	.57582	.12876

Table 2. A Significant Threshold value of an Accuracy rate comparison of KNN and DT Algorithm.

Independent Sample Test									
	Levene's Test for Equality of Variances		T-test for Equality of Means						
	F	Sig.	T	Df	Sig. (2-Tailed)	Mean Difference	Std. Error Differences	95% Confidence Interval of the Difference	
								Lower	Upper
Equal Variances assumed	.879	.354	9.81	38	.000	1.605	.163	1.27	1.936
Equal Variances not assumed			9.81	35.9	.000	1.605	.163	1.27	1.937

- In the present work, K Nearest Neighbor (KNN) is compared with Decision Tree (DT) , it depicts that the proposed algorithm gives more accuracy then compared algorithm.

DISCUSSION AND CONCLUSION

- Based on T-test Statistical analysis, the significance value of p=0.000 (independent sample T - test p<0.05) is obtained and shows that there is a statistical significant difference between the group 1 and group 2.
- In comparison of accuracy, K Nearest Neighbor (KNN) achieved 85.66% , outperforming the Decision Tree (DT) which achieved 84.05%, with an independent sample test indicating a significant difference in accuracy between the (p<0.05), with KNN achieving notably higher accuracy.
- The future scope for fraudulent scam call detection is likely to involve the integration of advanced technologies and the development of more sophisticated methods to combat evolving fraud techniques. The limitation of the research work is that it is supported for larger datasets, which is a restriction of the proposed work.
- The study presents fraudulent scam call detection and compares the accuracy of algorithms. The findings suggest that KNN algorithm (86%) is a better choice for scam call detection compared to Decision Tree algorithm (84%).

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